Built-in Functions

```
are functions that are globally available for use, without importing or
configuring
```

| Examples for Built-in Functions

https://docs.python.org/3/library/functions.html

		Built-in Functions		
abs()	delattr()	hash()	memoryview()	set()
all()	dict()	help()	min()	setattr()
any()	dir()	hex()	next()	slice()
ascii()	divmod()	id()	object()	sorted()
bin()	enumerate()	input()	oct()	staticmethod()
bool()	eval()	int()	open()	str()
breakpoint()	exec()	isinstance()	ord()	sum()
bytearray()	filter()	issubclass()	pow()	super()
bytes()	float()	iter()	print()	tuple()
callable()	format()	len()	property()	type()
chr()	frozenset()	list()	range()	vars()
classmethod()	getattr()	locals()	repr()	zip()
compile()	globals()	map()	reversed()	__import__()
complex()	hasattr()	max()	round()	

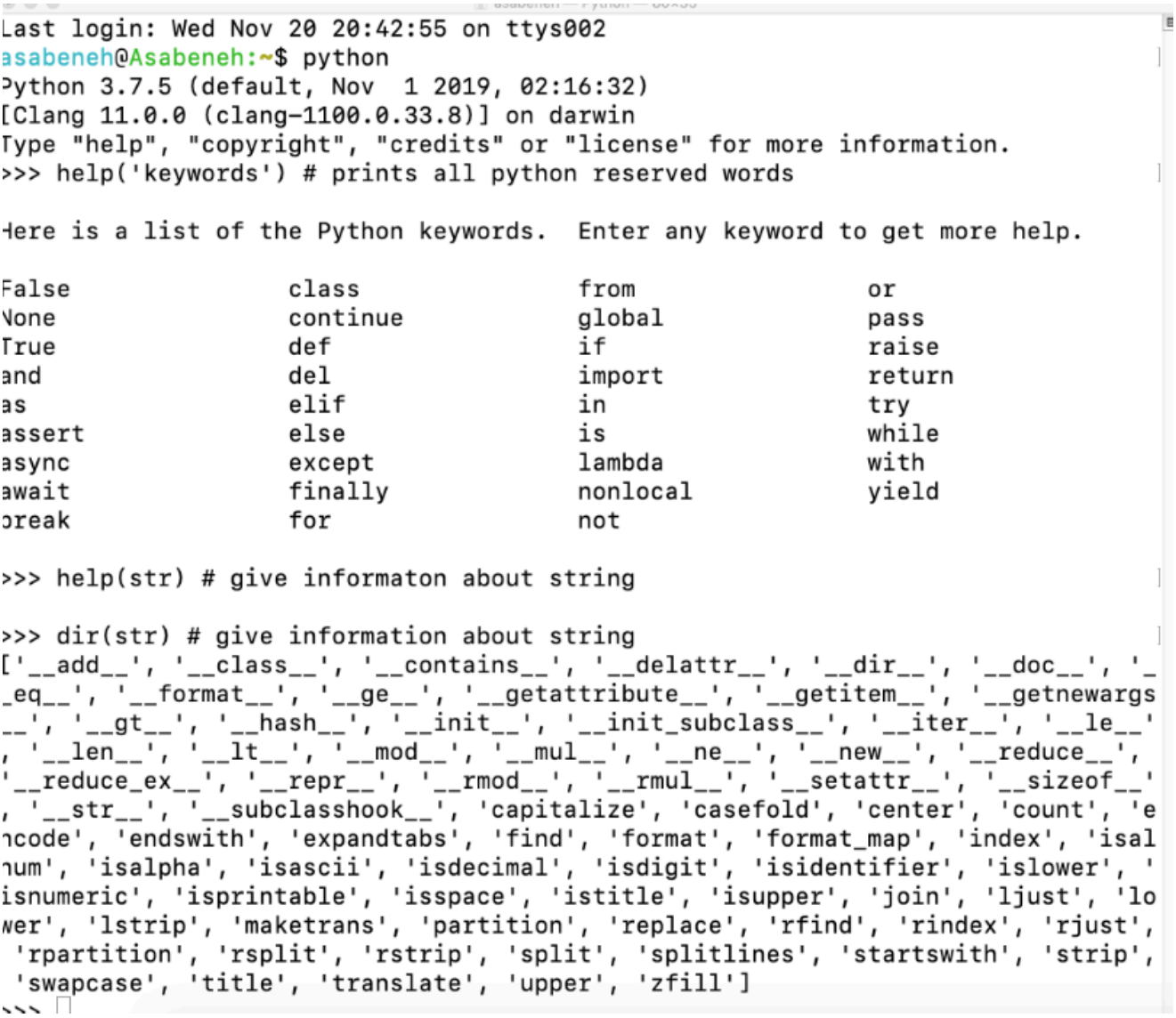
Additional

1. help() -> shows the documentation of module, function, class, or object
2. dir() -> shows available attributes or methods

| Reserved Words

```
are words with fixed meaning, cannot be redefined
```

Examples



```
Last login: Wed Nov 20 20:42:55 on ttys002
asabeneh@Asabeneh:~$ python
Python 3.7.5 (default, Nov  1 2019, 02:16:32)
[Clang 11.0.0 (clang-1100.0.33.8)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>> help('keywords') # prints all python reserved words

Here is a list of the Python keywords.  Enter any keyword to get more help.

False               class               from                or
None                continue            global              pass
True                def                 if                  raise
and                 del                 import              return
as                  elif                in                  try
assert              else                is                  while
async               except              lambda              with
await               finally             nonlocal            yield
break               for                 not

>>> help(str) # give informaton about string

>>> dir(str) # give information about string
['__add__', '__class__', '__contains__', '__delattr__', '__dir__', '__doc__', '_
_eq__', '__format__', '__ge__', '__getattribute__', '__getitem__', '__getnewargs
__', '__gt__', '__hash__', '__init__', '__init_subclass__', '__iter__', '__le__'
, '__len__', '__lt__', '__mod__', '__mul__', '__ne__', '__new__', '__reduce__',
'__reduce_ex__', '__repr__', '__rmod__', '__rmul__', '__setattr__', '__sizeof__'
, '__str__', '__subclasshook__', 'capitalize', 'casefold', 'center', 'count', 'e
ncode', 'endswith', 'expandtabs', 'find', 'format', 'format_map', 'index', 'isal
num', 'isalpha', 'isascii', 'isdecimal', 'isdigit', 'isidentifier', 'islower', '
isnumeric', 'isprintable', 'isspace', 'istitle', 'isupper', 'join', 'ljust', 'lo
wer', 'lstrip', 'maketrans', 'partition', 'replace', 'rfind', 'rindex', 'rjust',
 'rpartition', 'rsplit', 'rstrip', 'split', 'splitlines', 'startswith', 'strip',
 'swapcase', 'title', 'translate', 'upper', 'zfill']
>>>
```

Variables

```
store data in a memory, case sensitive, refer to memory addresses
```

| Naming Rule

1. Mnemonic variables are recommended, easily remembered and associated
2. No number at the beginning
3. No special character or hypen (-), only alphanumeric and underscores
4. Start with __ if the name are included in reserved words
5. Snake case (your_name_is) is recommended

| Variable Declaration

```
to assign data type to a variable
```

= -> assignment operator
assigning means storing data in the variable

| Arguments

```
values which can be passed or put inside function()
```

| Concatenation of Variables

```
use , or f"{}"
```

Data Types

| Checking Data Types and Casting

Checking by function type() and casting by int(), float(), str(), list(), set()